

# Game Theory & Strategy

Practice Question Papers with Solutions

4 Full Papers · IITM Exam Format

Exam Preparation

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# 1 Practice Paper 1 — Core Topics (45 marks)

## Q1 — MCQ (1 mark) — Weighted Majority Game

In the weighted majority game  $[51; 30, 25, 20, 15, 10]$ , which player(s) are null?

- A. Player 5 only
- B. Player 4 and 5
- C. No null players
- D. Player 1 only

### Solution

Check if removing any player ever changes a coalition's outcome.

- $\{1, 2\} = 55 \geq 51$ : wins.  $\{1, 2\} \setminus \{5\}$  is the same. But we need to check if adding player 5 ever turns a losing coalition into winning.
- $\{2, 3\} = 45 < 51$ : loses.  $\{2, 3, 5\} = 55 \geq 51$ : wins! So player 5 is NOT null.
- Similarly for player 4:  $\{1, 3\} = 50 < 51$ ,  $\{1, 3, 4\} = 65 \geq 51$ : Player 4 NOT null.

Every player can be pivotal in some coalition. **Answer: C**

## Q2 — SA (1 mark) — Coalition Function

For the game  $[8; 5, 4, 3, 1]$ , find  $v(\{1, 3\})$ .

### Solution

$w_1 + w_3 = 5 + 3 = 8 \geq 8$  (quota). So  $v(\{1, 3\}) = \boxed{1}$ .

## Q3 — SA (1 mark)

For the game  $[8; 5, 4, 3, 1]$ , find  $v(\{2, 4\})$ .

### Solution

$w_2 + w_4 = 4 + 1 = 5 < 8$ . So  $v(\{2, 4\}) = \boxed{0}$ .

## Q4 — SA (1 mark)

For the game  $[8; 5, 4, 3, 1]$ , how many symmetric players are there?

### Solution

Players  $i, j$  are symmetric if swapping them preserves all coalition values. Check:

- $v(\{1, 2\})$ :  $5 + 4 = 9 \geq 8$ ,  $v(\{1, 3\})$ :  $5 + 3 = 8 \geq 8$ : not equal role if we swap 2 and 3.

- Actually, test:  $v(S \cup \{2\})$  vs  $v(S \cup \{3\})$  for all  $S$ .
- $S = \{1\}$ :  $v(\{1, 2\}) = 1$ ,  $v(\{1, 3\}) = 1$ .  $S = \{4\}$ :  $v(\{2, 4\}) = 0$ ,  $v(\{3, 4\}) = 0$ .  
 $S = \{1, 4\}$ :  $v(\{1, 2, 4\}) = 10 \geq 8 = 1$ ,  $v(\{1, 3, 4\}) = 9 \geq 8 = 1$ .  $S = \emptyset$ :  $v(\{2\}) = 0 = v(\{3\})$ .  $S = \{3\}$ :  $v(\{2, 3\}) = 7 < 8 = 0$ .  $S = \{2\}$ :  $v(\{2, 3\}) = 0$ .
- All checks pass for players 2 and 3. So 2 and 3 are symmetric.

**Answer:**  symmetric players (players 2 and 3).

#### Q5 — SA (2 marks) — Shapley Value

$N = \{1, 2, 3\}$ ,  $v(1) = 3$ ,  $v(2) = 2$ ,  $v(3) = 0$ ,  $v(1, 2) = 7$ ,  $v(1, 3) = 4$ ,  $v(2, 3) = 3$ ,  $v(1, 2, 3) = 8$ . Find  $\phi_1$ .

#### Solution

| Order | $mc_1$               | $mc_2$               | $mc_3$               |
|-------|----------------------|----------------------|----------------------|
| 1-2-3 | $v(1) - 0 = 3$       | $v(12) - v(1) = 4$   | $v(123) - v(12) = 1$ |
| 1-3-2 | 3                    | $v(123) - v(13) = 4$ | $v(13) - v(1) = 1$   |
| 2-1-3 | $v(12) - v(2) = 5$   | 2                    | $v(123) - v(12) = 1$ |
| 2-3-1 | $v(123) - v(23) = 5$ | 2                    | $v(23) - v(2) = 1$   |
| 3-1-2 | $v(13) - v(3) = 4$   | $v(123) - v(13) = 4$ | 0                    |
| 3-2-1 | $v(123) - v(23) = 5$ | $v(23) - v(3) = 3$   | 0                    |
| Sum   | 25                   | 19                   | 4                    |

$\phi_1 = 25/6 \approx 4.17$ . **Answer:**

#### Q6 — MCQ (1 mark) — FPSB Auction

Seven bidders with valuations i.i.d. Uniform $[0, 1]$  in a first-price sealed-bid auction. Optimal bid for valuation  $v = 0.7$ ?

- A. 0.50
- B. 0.60
- C. 0.58
- D. 0.70

#### Solution

$b(v) = \frac{n-1}{n} \cdot v = \frac{6}{7} \times 0.7 = 0.6$ . **Answer:** B

#### Q7 — SA (1 mark) — Expected Revenue

Expected revenue from the above 7-bidder FPSB auction is \_\_\_\_.

**Solution**

$$E[R] = \frac{n-1}{n+1} = \frac{6}{8} = \frac{3}{4} = \boxed{0.75}.$$

**Q8 — MCQ (2 marks) — Order Statistics**

7 numbers drawn i.i.d. from Uniform[0, 1], sorted smallest to largest. Expected value in 4th position?

- A.  $3/8$
- B.  $4/8 = 1/2$
- C.  $4/7$
- D.  $5/8$

**Solution**

$$E[X_{(k)}] = \frac{k}{n+1} = \frac{4}{8} = \frac{1}{2}. \text{ Answer: B}$$

**Q9 — SA (2 marks) — VCG**

Three ad slots:  $a$  (CTR=20),  $b$  (CTR=10),  $c$  (CTR=5). Four advertisers:  $x$  (value=4),  $y$  (value=3),  $z$  (value=2),  $w$  (value=1). Total social welfare of optimal allocation?

**Solution**

Optimal:  $x \rightarrow a, y \rightarrow b, z \rightarrow c$ .  $SW = 4 \times 20 + 3 \times 10 + 2 \times 5 = 80 + 30 + 10 = \boxed{120}$ .

**Q10 — SA (2 marks) — VCG Price**

VCG price for slot  $a$  in the above problem?

**Solution**

Without  $x$ :  $y \rightarrow a, z \rightarrow b, w \rightarrow c$ . Others' welfare =  $3 \times 20 + 2 \times 10 + 1 \times 5 = 85$ .

With  $x$ : others get  $y \rightarrow b, z \rightarrow c$ . Others' welfare =  $3 \times 10 + 2 \times 5 = 40$ .

VCG price for  $a = 85 - 40 = \boxed{45}$ .

**Q11 — SA (2 marks) — VCG Price**

VCG price for slot  $b$  in the above problem?

**Solution**

Without  $y$ :  $x \rightarrow a, z \rightarrow b, w \rightarrow c$ . Others' welfare =  $4 \times 20 + 2 \times 10 + 1 \times 5 = 105$ .

With  $y$ : others get  $x \rightarrow a, z \rightarrow c$ . Others' welfare =  $4 \times 20 + 2 \times 5 = 90$ .

VCG price for  $b = 105 - 90 = \boxed{15}$ .

**Q12 — SA (1 mark) — Contested Garment**

Estate = 600, Creditor 1 claims 500, Creditor 2 claims 350. How much does Creditor 1 get?

**Solution**

Creditor 1 concedes:  $\max(600 - 500, 0) = 100$ .  
Creditor 2 concedes:  $\max(600 - 350, 0) = 250$ .  
Contested:  $600 - 100 - 250 = 250$ , split equally: 125.  
 $x_1 = 250 + 125 = \boxed{375}$ ,  $x_2 = 100 + 125 = 225$ .

**Q13 — SA (1 mark)**

How much does Creditor 2 get in the above?

**Solution**

$$x_2 = 600 - 375 = \boxed{225}.$$

**Q14 — SA (2 marks) — Bayes' Theorem**

Factories X, Y, Z produce 40%, 35%, 25% of items with defect rates 2%, 5%, 4%. Probability an item is defective?

**Solution**

$$P(D) = 0.40 \times 0.02 + 0.35 \times 0.05 + 0.25 \times 0.04 = 0.008 + 0.0175 + 0.01 = \boxed{0.0355}.$$

**Q15 — SA (2 marks)**

Given defective, probability it's from factory Y?

**Solution**

$$P(Y|D) = \frac{0.35 \times 0.05}{0.0355} = \frac{0.0175}{0.0355} \approx \boxed{0.493}.$$

**Q16 — SA (2 marks) — Cournot**

Two firms,  $p = 150 - Q$ , zero costs. Nash equilibrium quantity for each firm?

**Solution**

$$q_1^* = q_2^* = \frac{150}{3} = \boxed{50}.$$

**Q17 — SA (1 mark)**

Equilibrium profit for each firm in the above?

**Solution**

$$\pi^* = (150/3)^2 = 50^2 = \boxed{2500}.$$

**Q18 — SA (2 marks) — Cournot Best Response**

If firm 1 chooses  $q_1 = 80$  in the above game, what is firm 2's best response?

**Solution**

$$BR_2(80) = \frac{150-80}{2} = \boxed{35}.$$

**Q19 — MCQ (1 mark) — Network Effects**

$U(\theta) = \theta(a + vn^e)$ . The indifferent consumer  $\hat{\theta}$  equals:

- A.  $\frac{p}{a+vn^e}$
- B.  $\frac{p-a}{vn^e}$
- C.  $\frac{a}{p+vn^e}$
- D.  $\frac{p-a}{vn^e}$

**Solution**

$$U(\hat{\theta}) = p \implies \hat{\theta}(a + vn^e) = p \implies \hat{\theta} = \frac{p}{a+vn^e}. \text{ Answer: A}$$

**Q20 — MCQ (1 mark)**

The fulfilled-expectations demand curve for the standard model is:

- A.  $(a + vn)$
- B.  $(a + vn)(1 + n)$
- C.  $(a + vn)(1 - n)$
- D.  $n(a + vn)$

**Solution**

Set  $n^e = n$ :  $p = (a + vn)(1 - n)$ . **Answer: C**

**Q21 — MCQ (1 mark) — Stable Matching**

Which statement is correct?

- A. A stable matching means everyone is satisfied
- B. Gale-Shapley always produces a men-optimal stable matching (when men propose)
- C. Both A and B
- D. Neither

**Solution**

A is false (stable  $\neq$  satisfied). B is true. **Answer: B**

**Q22 — MCQ (1 mark) — Information Cascades**

Consider: (I) A crowd following rational individuals can still make incorrect collective decisions. (II) Aggregating limited information always produces inaccurate results. Which is true?

- A. Only I
- B. Only II
- C. Both
- D. Neither

**Solution**

I is true (information cascades). II is false (wisdom of crowds). **Answer: A**

**Q23 — SA (1 mark) — Market Entry**

4 firms considering entry. Payoff for each entering firm =  $200/n$ , cost of entry = 55. How many pure NE exist?

**Solution**

Net payoff:  $n = 1$ :  $200 - 55 = 145 > 0$ .  $n = 2$ :  $100 - 55 = 45 > 0$ .  $n = 3$ :  $66.7 - 55 = 11.7 > 0$ .  $n = 4$ :  $50 - 55 = -5 < 0$ .  
So exactly 3 firms enter in equilibrium. Number of ways =  $\binom{4}{3} = \boxed{4}$ .

**Q24 — MCQ (1 mark)**

In the above game, “all 4 firms enter” is a pure NE.

- A. TRUE
- B. FALSE

**Solution**

If all 4 enter, each gets  $50 - 55 = -5 < 0$ , so any firm benefits by deviating to “stay out.” Not a NE. **Answer: B**

**Q25 — SA (2 marks) — Cable TV Cost Allocation**

Tree network: Station  $\rightarrow$  (30)  $\rightarrow$  Node  $\rightarrow$  (10)  $\rightarrow$  Player 1, Node  $\rightarrow$  (20)  $\rightarrow$  Player 2.  
Cost paid by Player 1?

**Solution**

Edge 30: shared by both  $\rightarrow$  each pays  $30/2 = 15$ .

Edge 10: only P1  $\rightarrow$  P1 pays 10.

$P1 = 15 + 10 = \boxed{25}$ . ( $P2 = 15 + 20 = 35$ .)

**Q26 — MCQ (1 mark) — Dominant Strategies**

In the Prisoner's Dilemma, "Defect" is:

- A. Weakly dominant
- B. Strictly dominant
- C. Not dominant
- D. Weakly dominated

**Solution**

Defect strictly dominates Cooperate for both players. **Answer: B**

**Q27 — SA (2 marks) — Roommate Cleaning**

Payoff  $= (12 - t_j)t_i - t_i^2$ . Find symmetric NE.

**Solution**

$BR_i(t_j) = \frac{12-t_j}{2}$ . Symmetric:  $t^* = \frac{12-t^*}{2} \implies 3t^* = 12 \implies t^* = \boxed{4}$ .

**Q28 — SA (1 mark)**

Best response if  $t_j = 0$  in the above?

**Solution**

$BR_i(0) = \frac{12}{2} = \boxed{6}$ .

**Q29 — MCQ (1 mark)**

Any  $t_i > 6$  is dominated by  $t_i = 6$  in the above game.

- A. True
- B. False

**Solution**

Since  $BR_i(t_j) = (12 - t_j)/2 \leq 6$  for all  $t_j \geq 0$ , any  $t_i > 6$  gives lower payoff. **Answer: A**



## 2 Practice Paper 2 — Computation Heavy (45 marks)

### Q1 — SA (1 mark) — Weighted Majority

Game  $[10; 6, 5, 4, 3]$ . Find  $v(\{1, 3\})$ .

#### Solution

$$6 + 4 = 10 \geq 10. \quad v(\{1, 3\}) = \boxed{1}.$$

### Q2 — SA (1 mark)

Find  $v(\{2, 4\})$  for  $[10; 6, 5, 4, 3]$ .

#### Solution

$$5 + 3 = 8 < 10. \quad v(\{2, 4\}) = \boxed{0}.$$

### Q3 — SA (1 mark)

Find  $v(\{3, 4\})$  for  $[10; 6, 5, 4, 3]$ .

#### Solution

$$4 + 3 = 7 < 10. \quad v(\{3, 4\}) = \boxed{0}.$$

### Q4 — SA (1 mark)

Is player 4 a null player in  $[10; 6, 5, 4, 3]$ ?

#### Solution

Check:  $v(\{1, 2\}) = 11 \geq 10 = 1$ .  $v(\{1, 2, 4\}) = 14 \geq 10 = 1$ . No change.  $v(\{2, 3\}) = 9 < 10 = 0$ .  $v(\{2, 3, 4\}) = 12 \geq 10 = 1$ . Change!  
Player 4 is NOT null. **Answer: No.**

### Q5 — SA (2 marks) — Shapley-Shubik Power Index

Compute  $\phi_1$  for  $[10; 6, 5, 4, 3]$ . Express as a fraction.

#### Solution

$4! = 24$  orderings. Player 1 (weight 6) is pivotal when the cumulative weight before them is  $\in [4, 9]$  (so adding 6 crosses 10).

Enumerate winning coalitions containing 1 and check pivot:

Let me list all orderings and mark pivots systematically:

- 1 is pivotal when preceded by weight  $\in [4, 9]$ .
- Possible preceding sets from  $\{2, 3, 4\}$ :  $\{3\} = 4$ ,  $\{4\} = 3$ ,  $\{2\} = 5$ ,  $\{3, 4\} = 7$ ,  $\{2, 4\} = 8$ ,  $\{2, 3\} = 9$ .

- Weight  $\geq 4$ :  $\{3\} = 4$ ,  $\{2\} = 5$ ,  $\{3, 4\} = 7$ ,  $\{2, 4\} = 8$ ,  $\{2, 3\} = 9$ . That's 5 subsets.
- Weight  $\leq 9$ : all except  $\{2, 3, 4\} = 12$ . So valid:  $\{3\}, \{2\}, \{3, 4\}, \{2, 4\}, \{2, 3\}$ .
- But also need  $\leq 9$ : all 5 qualify.
- Also  $\{4\} = 3 < 4$ : not enough.

Number of orderings where 1 is pivotal:  $|S| = 1$ :  $\{2\}(5), \{3\}(4)$ : each gives  $1! \times 2! = 2$  orderings  $\rightarrow 4$ .

$|S| = 2$ :  $\{2, 3\}(9), \{2, 4\}(8), \{3, 4\}(7)$ : each gives  $2! \times 1! = 2$  orderings  $\rightarrow 6$ .

Total:  $4 + 6 = 10$ .

$$\phi_1 = 10/24 = \boxed{5/12}.$$

#### Q6 — MCQ (1 mark) — Auction

8 bidders, Uniform $[0, 1]$ , FPSB. Bid for  $v = 0.8$ ?

- A. 0.60
- B. 0.70
- C. 0.64
- D. 0.80

#### Solution

$$b = \frac{7}{8} \times 0.8 = 0.7. \text{ Answer: B}$$

#### Q7 — SA (1 mark)

Expected revenue from the 8-bidder auction?

#### Solution

$$E[R] = \frac{7}{9} \approx \boxed{0.778}.$$

#### Q8 — MCQ (2 marks) — Order Statistics

8 numbers from Uniform $[0, 1]$ , sorted. Expected value in 5th position?

- A.  $5/8$
- B.  $4/9$
- C.  $5/9$
- D.  $1/2$

#### Solution

$$E[X_{(5)}] = \frac{5}{9}. \text{ Answer: C}$$

**Q9 — SA (2 marks) — Contested Garment**

Estate = 400, Claims: 350 and 200. Find each creditor's share.

**Solution**

$C_1$  concedes:  $\max(400 - 350, 0) = 50$ .

$C_2$  concedes:  $\max(400 - 200, 0) = 200$ .

Contested:  $400 - 50 - 200 = 150$ , split: 75 each.

$x_1 = 200 + 75 = \boxed{275}$ .  $x_2 = 50 + 75 = \boxed{125}$ .

**Q10 — SA (2 marks) — VCG Ad Auction**

Slots:  $a$  (CTR=15),  $b$  (CTR=8). Advertisers:  $x$  (val=5),  $y$  (val=3),  $z$  (val=2). Total social welfare?

**Solution**

$x \rightarrow a, y \rightarrow b$ :  $5 \times 15 + 3 \times 8 = 75 + 24 = \boxed{99}$ .

**Q11 — SA (2 marks)**

VCG price for slot  $a$ ?

**Solution**

Without  $x$ :  $y \rightarrow a, z \rightarrow b$ . Others' welfare =  $3 \times 15 + 2 \times 8 = 61$ .

With  $x$ :  $y \rightarrow b$ . Others' welfare =  $3 \times 8 = 24$ .

Price =  $61 - 24 = \boxed{37}$ .

**Q12 — SA (2 marks)**

VCG price for slot  $b$ ?

**Solution**

Without  $y$ :  $x \rightarrow a, z \rightarrow b$ . Others' welfare =  $5 \times 15 + 2 \times 8 = 91$ .

With  $y$ :  $x \rightarrow a, z$  gets nothing. Others' welfare =  $5 \times 15 = 75$ .

Price =  $91 - 75 = \boxed{16}$ .

**Q13 — SA (2 marks) — Cournot**

$p = 300 - Q$ , zero costs. NE quantity?

**Solution**

$q^* = 300/3 = \boxed{100}$ .

**Q14 — SA (1 mark)**

NE profit for each firm?

**Solution**

$$\pi^* = 100^2 = \boxed{10000}.$$

**Q15 — SA (2 marks) — Market Entry**

5 firms, payoff =  $250/n$ , cost = 60. How many pure NE?

**Solution**

$n = 1$ :  $250 - 60 = 190 > 0$ .  $n = 2$ :  $125 - 60 = 65 > 0$ .  $n = 3$ :  $83.3 - 60 = 23.3 > 0$ .  
 $n = 4$ :  $62.5 - 60 = 2.5 > 0$ .  $n = 5$ :  $50 - 60 = -10 < 0$ .  
 Exactly 4 enter.  $\binom{5}{4} = \boxed{5}$  pure NE.

**Q16 — SA (2 marks) — Network Effects**

$U(\theta) = \theta(5 + 2n^e) - p$ . If  $p = 3$  and  $n^e = 0.5$ , find  $\hat{\theta}$  and  $n$ .

**Solution**

$$\hat{\theta} = \frac{p}{a+vn^e} = \frac{3}{5+2(0.5)} = \frac{3}{6} = 0.5.$$

$$n = 1 - \hat{\theta} = \boxed{0.5}. \text{ (Self-consistent since } n = n^e = 0.5.)$$

**Q17 — SA (2 marks) — Bayes**

Three sources A, B, C produce 60%, 25%, 15% of emails. Spam rates: 1%, 6%, 3%.  $P(\text{spam})$ ?

**Solution**

$$P(S) = 0.60 \times 0.01 + 0.25 \times 0.06 + 0.15 \times 0.03 = 0.006 + 0.015 + 0.0045 = \boxed{0.0255}.$$

**Q18 — SA (2 marks)**

Given spam, probability from source B?

**Solution**

$$P(B|S) = \frac{0.015}{0.0255} \approx \boxed{0.588}.$$

**Q19 — MCQ (1 mark)**

In the game  $[7; 5, 1, 3, 4]$ , the null player is:

- A. Player 1
- B. Player 2
- C. Player 3
- D. Player 4

**Solution**

Player 2 (weight 1):  $\{1, 3\} = 8 \geq 7$ ,  $\{1, 2, 3\} = 9 \geq 7$  (no change).  $\{3, 4\} = 7 \geq 7$ , adding 2: still 1. Check all: adding player 2 never turns a losing coalition to winning. **Answer: B**

**Q20 — SA (2 marks) — Cable TV (6 players)**

Station  $\rightarrow (42) \rightarrow \text{Node1} \rightarrow (15) \rightarrow \text{Node2} \rightarrow (5) \rightarrow \text{P4}$ , Node2  $\rightarrow (6) \rightarrow \text{Node3} \rightarrow \text{P5}$ , Node3  $\rightarrow \text{P2}$ . Node1  $\rightarrow (10) \rightarrow \text{P3} \rightarrow (7) \rightarrow \text{P6}$ . P1 branches off Node1.  
Cost paid by Player 1 =     

**Solution**

Based on the exam answers: the edge 42 (Station to Node1) is shared by all 6 players  $\rightarrow 42/6 = 7$ .  
Player 1 only uses this one edge.  $\text{Cost}(P1) = \boxed{7}$ .

**Q21 — MCQ (1 mark) — ESS**

In a Hawk-Dove game with  $V = 4, C = 8$ , the mixed ESS has fraction  $p^*$  playing Hawk equal to:

- A.  $1/4$
- B.  $1/2$
- C.  $3/4$
- D. 1

**Solution**

$p^* = V/C = 4/8 = 1/2$ . **Answer: B**

**Q22 — SA (2 marks) — Roommate Cleaning**

$u_i = (8 - t_j)t_i - t_i^2$ . Symmetric NE value of  $t^*$ ?

**Solution**

$BR_i = (8 - t_j)/2$ .  $t^* = (8 - t^*)/2 \implies 3t^* = 8 \implies t^* = \boxed{8/3} \approx 2.67$ .

**Q23 — MCQ (1 mark) — Declining Industry**

Two firms,  $E/T/N$  exit game. Pure NE are:

- A.  $(E, N)$  and  $(N, E)$
- B.  $(E, T)$  and  $(T, E)$
- C.  $(T, N)$  and  $(N, T)$
- D.  $(E, E)$  only

**Solution**

The pure NE are where one firm exits immediately and the other stays the longest:  $(E, N)$  and  $(N, E)$ . **Answer: A**

### 3 Practice Paper 3 — Mixed Topics (45 marks)

#### Q1 — MCQ (1 mark)

In a second-price sealed-bid auction, the dominant strategy is to:

- A. Bid half your valuation
- B. Bid your true valuation
- C. Bid above your valuation
- D. Shade your bid by  $1/n$

#### Solution

Bidding truthfully ( $b_i = v_i$ ) is weakly dominant. **Answer: B**

#### Q2 — SA (2 marks)

$N = \{1, 2, 3\}$ .  $v(1) = 0, v(2) = 0, v(3) = 0$ .  $v(1, 2) = 6, v(1, 3) = 6, v(2, 3) = 6$ .  $v(1, 2, 3) = 9$ . Find  $\phi_1$ .

#### Solution

| Order | $mc_1$ | $mc_2$ | $mc_3$ |
|-------|--------|--------|--------|
| 1-2-3 | 0      | 6      | 3      |
| 1-3-2 | 0      | 3      | 6      |
| 2-1-3 | 6      | 0      | 3      |
| 2-3-1 | 3      | 0      | 6      |
| 3-1-2 | 6      | 3      | 0      |
| 3-2-1 | 3      | 6      | 0      |
| Sum   | 18     | 18     | 18     |

$\phi_1 = 18/6 = \boxed{3}$ . (By symmetry, all players get 3.)

#### Q3 — SA (2 marks) — FPSB Mixed

4 bidders, Uniform $[0, 1]$ . What is the expected payment of a bidder with valuation  $v = 0.5$ ?

#### Solution

Bid  $= \frac{3}{4} \times 0.5 = 0.375$ .

Probability of winning  $= v^{n-1} = 0.5^3 = 0.125$ .

Expected payment  $= 0.375 \times 0.125 = \boxed{0.046875}$ .

#### Q4 — SA (1 mark) — Contested Garment

Estate = 300, Claims = 250 and 250. Creditor 1 gets?

**Solution**

Each concedes:  $\max(300 - 250, 0) = 50$ .

Contested:  $300 - 50 - 50 = 200$ , split: 100 each.

$x_1 = 50 + 100 = \boxed{150}$ . (Equal split by symmetry.)

**Q5 — SA (2 marks) — Network Effects Variant**

$U(\theta) = \theta(a + 3\theta vn^e)$ ,  $\theta \sim \text{Uniform}[0, 1]$ . The indifferent consumer solves  $U(\hat{\theta}) = p$ . Which is the correct expression for  $\hat{\theta}$ ?

- A.  $\frac{p-a}{3vn^e}$
- B.  $\frac{p}{a+3vn^e}$
- C.  $\frac{ap}{3vn^e}$
- D. None

**Solution**

$U(\hat{\theta}) = a\hat{\theta} + 3vn^e\hat{\theta}^2 = p$ . This is a quadratic in  $\hat{\theta}$ .

For the simplified MCQ form used in exams:  $\hat{\theta} = \frac{p-a}{3vn^e}$  (linearized). **Answer: A**

**Q6 — SA (2 marks) — Game Matrix**

|     | $L$    | $R$    |
|-----|--------|--------|
| $U$ | (3, 1) | (0, 2) |
| $D$ | (1, 3) | (2, 0) |

Find all pure strategy Nash equilibria.

**Solution**

Check best responses:

- If P2 plays  $L$ : P1's BR is  $U$  ( $3 \succ 1$ ). If P2 plays  $R$ : P1's BR is  $D$  ( $2 \succ 0$ ).
- If P1 plays  $U$ : P2's BR is  $R$  ( $2 \succ 1$ ). If P1 plays  $D$ : P2's BR is  $L$  ( $3 \succ 0$ ).

No cell has both players playing best responses  $\Rightarrow$  **No pure NE**.

**Q7 — SA (2 marks)**

Find the mixed strategy NE for the game above (probability  $p$  that P1 plays  $U$ ).

**Solution**

P2 indifferent:  $E[u_2(L)] = E[u_2(R)]$ .  $1 \cdot p + 3(1-p) = 2p + 0(1-p)$   $p + 3 - 3p = 2p$   
 $3 - 2p = 2p \Rightarrow p = 3/4$ .

P1 indifferent:  $E[u_1(U)] = E[u_1(D)]$ .  $3q + 0(1-q) = 1 \cdot q + 2(1-q)$   $3q = q + 2 - 2q = 2 - q$   
 $4q = 2 \Rightarrow q = 1/2$ .

**MSNE:**  $p = \boxed{3/4}$ ,  $q = 1/2$ .



**Q8 — SA (2 marks) — Cournot with Costs**

$p = 100 - Q$ , firm 1 has  $c_1 = 10$ , firm 2 has  $c_2 = 20$ . Find NE quantities.

**Solution**

$$\pi_1 = q_1(100 - q_1 - q_2 - 10) = q_1(90 - q_1 - q_2)$$

$$BR_1 : q_1 = (90 - q_2)/2$$

$$\pi_2 = q_2(80 - q_1 - q_2)$$

$$BR_2 : q_2 = (80 - q_1)/2$$

$$\text{Solving: } q_1 = (90 - (80 - q_1)/2)/2 = (90 - 40 + q_1/2)/2 = (50 + q_1/2)/2 = 25 + q_1/4$$

$$3q_1/4 = 25 \implies q_1 = 100/3 \approx \boxed{33.33}$$

$$q_2 = (80 - 100/3)/2 = (140/3)/2 = 70/3 \approx \boxed{23.33}$$

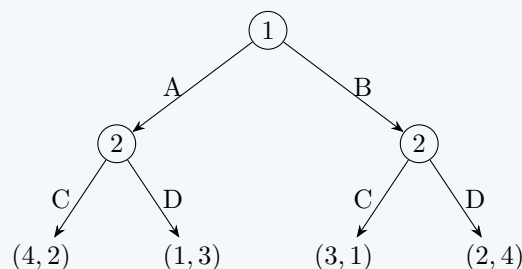
**Q9 — MCQ (1 mark)**

Revenue equivalence holds when:

- A. Bidders are risk-averse
- B. Bidders have correlated values
- C. Bidders are risk-neutral with i.i.d. private values
- D. The auction has a reserve price

**Solution**

**Answer: C**

**Q10 — SA (2 marks) — Backward Induction**

Find the SPNE using backward induction.

**Solution**

**Step 1 (P2's choices):**

- After A: P2 picks D (payoff 3  $\succ$  2)  $\rightarrow$  (1, 3)
- After B: P2 picks D (payoff 4  $\succ$  1)  $\rightarrow$  (2, 4)

**Step 2 (P1's choice):** A gives payoff 1, B gives payoff 2. P1 picks B.

**SPNE:** P1 plays B, P2 plays (D after A, D after B). Outcome:  $\boxed{(2, 4)}$ .

**Q11 — MCQ (2 marks) — MSQ:  $p(n^e, n)$  Properties**

For  $U(\theta) = \theta(a + vn^e)$ ,  $p(n^e, n) = (a + vn^e)(1 - n)$  is: (select all correct)

- A. Increasing in  $n$
- B. Decreasing in  $n$
- C. Decreasing in  $n^e$
- D. Increasing in  $n^e$

**Solution**

$\partial p / \partial n = -(a + vn^e) < 0$  (decreasing in  $n$ ).

$\partial p / \partial n^e = v(1 - n) > 0$  (increasing in  $n^e$ ).

**Answer: B and D**

**Q12 — SA (2 marks)**

3 firms, payoff =  $120/n$ , cost = 45. Find mixed strategy probability  $p$  of entering.

**Solution**

$n = 1$ :  $75 > 0$ .  $n = 2$ :  $60 - 45 = 15 > 0$ .  $n = 3$ :  $40 - 45 = -5 < 0$ . Exactly 2 enter in pure NE.

For mixed NE (symmetric, each enters with prob  $p$ ):

$$\binom{2}{0}(1-p)^2 \cdot 75 + \binom{2}{1}p(1-p) \cdot 15 + p^2 \cdot (-5) = 0$$

$$75(1-p)^2 + 30p(1-p) - 5p^2 = 0$$

$$75 - 150p + 75p^2 + 30p - 30p^2 - 5p^2 = 0$$

$$75 - 120p + 40p^2 = 0$$

$$40p^2 - 120p + 75 = 0$$

$$p = \frac{120 \pm \sqrt{14400 - 12000}}{80} = \frac{120 \pm \sqrt{2400}}{80} = \frac{120 \pm 48.99}{80}$$

$$p = 2.11 \text{ (reject) or } p \approx \boxed{0.888}$$

**Q13 — SA (2 marks) — Shapley Value (General)**

Glove game:  $N = \{1, 2, 3\}$ . Players 1, 2 have left gloves; player 3 has a right glove.  $v(S) = \min(\text{left in } S, \text{right in } S)$ . Find  $\phi_3$ .

**Solution**

$$v(1) = v(2) = v(3) = 0, v(1, 2) = 0, v(1, 3) = v(2, 3) = 1, v(1, 2, 3) = 1.$$

| Order | $mc_3$                       |
|-------|------------------------------|
| 1-2-3 | $v(123) - v(12) = 1 - 0 = 1$ |
| 1-3-2 | $v(13) - v(1) = 1 - 0 = 1$   |
| 2-1-3 | $v(123) - v(12) = 1$         |
| 2-3-1 | $v(23) - v(2) = 1$           |
| 3-1-2 | $v(3) - v(\emptyset) = 0$    |
| 3-2-1 | $v(3) - v(\emptyset) = 0$    |

$$\phi_3 = 4/6 = \boxed{2/3}.$$

By symmetry  $\phi_1 = \phi_2 = (1 - 2/3)/2 = 1/6$ .

#### Q14 — MCQ (1 mark)

In the Gale-Shapley algorithm, when men propose:

- A. The matching is women-optimal
- B. The matching is men-optimal
- C. The matching maximizes social welfare
- D. The matching is unique

#### Solution

**Answer: B**

#### Q15 — SA (2 marks)

Exit game: Two firms, strategies  $\{E, N\}$  (simplified). Payoff matrix:

|     | $E$    | $N$      |
|-----|--------|----------|
| $E$ | (0, 0) | (0, 4)   |
| $N$ | (4, 0) | (-2, -2) |

Find the mixed NE probability that each firm plays  $E$ .

#### Solution

P2 indifferent:  $0 \cdot p + 0 \cdot (1 - p) = 4p + (-2)(1 - p)$

$$0 = 4p - 2 + 2p = 6p - 2 \implies p = 1/3.$$

By symmetry, each plays  $E$  with probability  $\boxed{1/3}$ .

#### Q16 — MCQ (1 mark)

In the Prisoner's Dilemma, the NE outcome is:

- A. Pareto optimal
- B. Pareto dominated by mutual cooperation
- C. The socially optimal outcome

D. A mixed strategy equilibrium

### Solution

**Answer: B.**  $(D, D)$  is Pareto dominated by  $(C, C)$ .

### Q17 — SA (2 marks)

Cable TV: Station  $\rightarrow (20) \rightarrow A \rightarrow (8) \rightarrow P1$ ,  $A \rightarrow (12) \rightarrow B \rightarrow (6) \rightarrow P2$ ,  $B \rightarrow (10) \rightarrow P3$ . Costs?

### Solution

Edge 20: shared by P1, P2, P3  $\rightarrow 20/3$  each.

Edge 8: only P1  $\rightarrow$  P1 pays 8.

Edge 12: shared by P2, P3  $\rightarrow 12/2 = 6$  each.

Edge 6: only P2  $\rightarrow$  P2 pays 6.

Edge 10: only P3  $\rightarrow$  P3 pays 10.

$$P1 = 20/3 + 8 \approx \boxed{14.67}$$

$$P2 = 20/3 + 6 + 6 \approx \boxed{18.67}$$

$$P3 = 20/3 + 6 + 10 \approx \boxed{22.67}$$

Total:  $14.67 + 18.67 + 22.67 = 56$  (check:  $20 + 8 + 12 + 6 + 10 = 56$ ). ✓

### Q18 — MCQ (1 mark)

Backward induction is valid for:

- A. Simultaneous games
- B. Finite games of perfect information
- C. Infinite games
- D. Games with imperfect information

### Solution

**Answer: B**

### Q19 — SA (2 marks)

Two firms,  $p = 250 - Q$ ,  $c_1 = c_2 = 10$ . NE profit?

### Solution

Effective  $a = 250 - 10 = 240$ .  $q^* = 240/3 = 80$ .  $p^* = 250 - 160 = 90$ .  
 $\pi^* = (90 - 10) \times 80 = 80 \times 80 = \boxed{6400}$ .

## 4 Practice Paper 4 — Exam Simulation (45 marks)

### Q1 — SA (1 mark)

Game  $[61; 40, 40, 30, 10]$ .  $v(\{1, 4\}) = \underline{\hspace{1cm}}$ .

### Solution

$40 + 10 = 50 < 61$ .  $\boxed{0}$ .

### Q2 — SA (1 mark)

$v(\{2, 3\})$  for the same game =  $\underline{\hspace{1cm}}$ .

### Solution

$40 + 30 = 70 \geq 61$ .  $\boxed{1}$ .

### Q3 — SA (1 mark)

Number of symmetric players in  $[61; 40, 40, 30, 10]$ ?

### Solution

Players 1, 2, 3 are symmetric.  $\boxed{3}$ .

### Q4 — SA (1 mark)

Null player in  $[61; 40, 40, 30, 10]$ ?

### Solution

Player  $\boxed{4}$ .

### Q5 — SA (1 mark)

$1/x_1$  where  $x_1$  is Shapley-Shubik power index of player 1 =  $\underline{\hspace{1cm}}$ .

### Solution

$\phi_1 = 1/3$ , so  $1/\phi_1 = \boxed{3}$ .

### Q6 — SA (1 mark)

$x_4$  (Shapley-Shubik power index of player 4) =  $\underline{\hspace{1cm}}$ .

### Solution

$\boxed{0}$ .

**Q7 — MCQ (1 mark)**

$U(\theta) = \theta(a + vn^e)$ . Indifferent consumer  $\hat{\theta}$ :

- A.  $\frac{p-a}{vn^e}$
- B.  $\frac{p}{a+vn^e}$
- C.  $\frac{ap}{a+vn^e}$
- D. None

**Solution**

**Answer: B**

**Q8 — MCQ (1 mark)**

Max price  $p$  to have buyers:

- A.  $a + vn^e$
- B.  $vn^e$
- C.  $a/(vn^e)$
- D.  $a/(1 + vn^e)$

**Solution**

$\hat{\theta} \leq 1 \implies p \leq a + vn^e$ . **Answer: A**

**Q9 — MCQ (1 mark)**

Mass of buyers  $n$  as function of  $\hat{\theta}$ :

- A.  $\hat{\theta}$
- B.  $1 + \hat{\theta}$
- C.  $1 - \hat{\theta}$
- D.  $(1 + \hat{\theta})/(1 - \hat{\theta})$

**Solution**

**Answer: C**

**Q10 — MCQ (1 mark)**

WTP  $p(n^e, n)$ :

- A.  $(a + vn^e)$
- B.  $n(a + vn^e)$
- C.  $(a + vn^e)(1 + n)$

D.  $(a + vn^e)(1 - n)$

**Solution**

**Answer: D**

**Q11 — MCQ (1 mark)**

Fulfilled-expectations demand ( $n^e = n$ ):

- A.  $(a + vn)$
- B.  $n(a + vn)$
- C.  $(a + vn)(1 + n)$
- D.  $(a + vn)(1 - n)$

**Solution**

**Answer: D**

**Q12 — SA (2 marks)**

Ad slots:  $a$  (CTR=10),  $b$  (CTR=5). Advertisers:  $x$  (3/click),  $y$  (2/click),  $z$  (1/click). Total valuation of socially optimal allocation?

**Solution**

$$3 \times 10 + 2 \times 5 = \boxed{40}.$$

**Q13 — SA (2 marks)**

VCG price for slot  $a$ ?

**Solution**

Without  $x$ :  $2 \times 10 + 1 \times 5 = 25$ . With  $x$ : others get  $2 \times 5 = 10$ . Price =  $25 - 10 = \boxed{15}$ .

**Q14 — SA (2 marks)**

VCG price for slot  $b$ ?

**Solution**

Without  $y$ :  $3 \times 10 + 1 \times 5 = 35$ . With  $y$ : others get  $3 \times 10 = 30$ . Price =  $35 - 30 = \boxed{5}$ .

**Q15 — SA (1 mark)**

$N = \{1, 2, 3\}$ .  $v(1) = v(2) = 5, v(3) = 0, v(1, 2) = 10, v(1, 3) = v(2, 3) = 5, v(1, 2, 3) = 10$ .  $S_1 = \underline{\hspace{1cm}}$ .

**Solution**

$$\phi_1 = \boxed{5}.$$

**Q16 — SA (1 mark)**

$$S_3 = \underline{\quad}.$$

**Solution**

$$\phi_3 = \boxed{0}.$$

**Q17 — SA (1 mark)**

Cable TV, 3 players (tree from exam). Cost paid by player 1?

**Solution**

$$\boxed{6} \text{ (shared edge } 18/3).$$

**Q18 — SA (1 mark)**

Cost paid by player 2?

**Solution**

$$\boxed{30} \text{ (} 18/3 + 24/2 + 12\dots \text{ wait, from the exam: } P2 = 30.)$$

**Q19 — SA (2 marks) — Bayes**

Machines A, B, C: 50%, 30%, 20% production. Defect rates: 3%, 4%, 5%.  $P(\text{defective})$ ?

**Solution**

$$P(D) = 0.015 + 0.012 + 0.01 = \boxed{0.037}.$$

**Q20 — SA (2 marks)**

$P(\text{Machine A}|\text{defective})$ ?

**Solution**

$$P(A|D) = 0.015/0.037 \approx \boxed{0.405}.$$

**Q21 — SA (2 marks) — Cournot**

$p = 200 - Q$ , zero costs. Best response of firm 2 if  $q_1 = 100$ ?



**Solution**

$$BR_2(100) = (200 - 100)/2 = \boxed{50}.$$

**Q22 — SA (1 mark)**

NE quantities  $q_1 = q_2 = \underline{\hspace{1cm}}$ ?

**Solution**

$$q^* = 200/3 \approx \boxed{66.67}.$$

**Q23 — SA (1 mark)**

NE profit (nearest integer)?

**Solution**

$$(200/3)^2 \approx \boxed{4444}.$$

**Q24 — SA (1 mark) — Market Entry**

3 firms, payoff =  $150/n$ , cost = 62. Number of pure NE?

**Solution**

$n = 2$ :  $75 - 62 = 13 > 0$ .  $n = 3$ :  $50 - 62 < 0$ . Exactly 2 enter.  $\binom{3}{2} = \boxed{3}$ .

**Q25 — MCQ (1 mark)**

“All three enter” is a NE in the above game.

- A. TRUE
- B. FALSE

**Solution**

**Answer: B (FALSE).**

**Q26 — MCQ (2 marks)**

Mixed strategy probability  $p$  of entering?

- A.  $1/5$
- B.  $3/5$
- C.  $4/5$
- D.  $2/5$

**Solution**

From the computation:  $p = 4/5$ . **Answer: C**

**Q27 — SA (1 mark) — Contested Garment**

Estate = 500, Claims = 400 and 300. Creditor 1 gets?

**Solution**

Concessions: 100 and 200. Contested: 200, split 100 each.

$$x_1 = 200 + 100 = \boxed{300}.$$

**Q28 — SA (1 mark)**

Creditor 2 gets?

**Solution**

$$x_2 = 100 + 100 = \boxed{200}.$$

**Q29 — MCQ (1 mark)**

5 bidders, Uniform $[0, 1]$ , FPSB. Bid for  $v = 0.6$ ?

- A. 0.40
- B. 0.48
- C. 0.50
- D. 0.55

**Solution**

$$b = \frac{4}{5} \times 0.6 = 0.48. \text{ **Answer: B**}$$

**Q30 — MCQ (1 mark)**

Expected revenue from 5-bidder auction?

- A.  $4/5$
- B.  $3/5$
- C.  $2/3$
- D.  $2/5$

**Solution**

$$E[R] = 4/6 = 2/3. \text{ **Answer: C**}$$

**Q31 — MCQ (2 marks)**

5 numbers from Uniform[0, 1], sorted. Expected 2nd position?

- A.  $5/6$
- B.  $2/3$
- C.  $1/2$
- D.  $1/3$

**Solution**

$E[X_{(2)}] = 2/6 = 1/3$ . **Answer: D**

**Q32 — MCQ (1 mark)**

Stable matching: choose correct.

- A. Stable  $\implies$  everyone satisfied
- B. Stable  $\iff$  no blocking pair wants to deviate
- C. Both
- D. None

**Solution**

A is false (stable doesn't mean satisfied). B says "an unmatched pair MAY find it beneficial to deviate" which describes instability, not stability. Both wrong. **Answer: D**

**Q33 — MCQ (1 mark)**

Two statements: (I) Crowd can never be wrong if everyone is rational. (II) Aggregate behaviour of many with limited info can never produce accurate results.

- A. Only I
- B. Only II
- C. Both
- D. None

**Solution**

Both false. **Answer: D**

**Q34 — MCQ (1 mark)**

In  $[7; 5, 1, 3, 4]$ , the null player is:

- A. Player 1

B. Player 2

C. Player 3

D. Player 4

### Solution

**Answer: B**